

Apply an Advanced AI Technique

MLflow Experiment Tracking

1. (Why they selected the technique)

We choose MLflow Experiment Tracking because it connects naturally to the MLOps engineering workflow. When you train machine learning models, it is very messy. You keep tweaking settings, and it is hard to remember which settings give the best score. MLflow acts like a digital lab notebook to remember everything automatically.

2. (How it was implemented)

We built an MLflow tracking block inside our pipeline script. Every time the Random Forest model runs, the script intercepts the model settings and performance scores. It automatically saves the hyperparameters as Parameters, saves the final validation metrics,

3.(What value it adds)

It removes all the guesswork. You can just look at the MLflow dashboard history sheet to see the exact settings that worked. It makes the whole capstone project highly structured, organized, and reproducible.

1. (Its limitation)

The main limitation is that MLflow does not fix bad code or clean dirty datasets for you, it only tracks what you tell it to track. If you run it locally on your computer, the tracking logs are stuck on your hard drive, so teammates cannot view your history sheet unless you deploy an expensive central web server to share it online.